

AP CSP | Lesson 08: Data Analysis & Evidence

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Chapter 2 Check: 12 original questions, 30 classroom marks. Q5 requires exactly TWO selections. Use the synthetic retained sample and stated analysis views. Show units, counts, evidence and limitations.

1. A model increases from 4 to 16 samples over the same time window while retaining 4 value levels. Which statement is supported? [1 mark]

A. Each sample now has 16 value levels. B. Measurements are taken at more times.
C. The original signal must be represented exactly. D. The bits needed per sample must quadruple.

Answer: _____

2. A school must archive numerical records so that every original bit can be recovered. Which requirement is essential? [1 mark]

A. Use the smallest file regardless of information loss. B. Delete unusual values before saving.
C. Use only a technique guaranteed to halve every file. D. Use lossless compression if the data are compressed.

Answer: _____

3. The synthetic survey contains a 120-minute response. It is unusual but has no confirmed error. What is the most defensible baseline treatment? [1 mark]

A. Retain and flag it; document any separate sensitivity comparison. B. Delete it because the mean becomes lower.
C. Replace it with the next-largest value without recording the change. D. Assume it describes every bus rider.

Answer: _____

4. The 13 retained records are sorted from shortest to longest time, with no values changed. Which statement must be true? [1 mark]

A. The mean decreases. B. Only the shortest records remain.
C. The same records contribute to the mean. D. The 120-minute record is removed.

Answer: _____

5. The retained sample has Walk mean 11.25 minutes (n=4), Bike mean 17.67 (n=3), and Bus mean 47.17 (n=6). Which TWO claims are supported? Select exactly TWO answers. [1 mark]

A. Bus caused each student to travel longer. B. Bus has the highest mean in this retained sample.
C. The three groups contain equal numbers of responses. D. The retained sample contains 13 responses in total.

Answer: _____

6. The retained Walk times are 8, 10, 12 and 15 minutes. What is their mean? [1 mark]

A. 10 B. 11 C. 11.25 D. 45

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A recorder stores 40 samples using 3 bits per sample, with no overhead or compression.

- (a) Find the sample payload in bits and bytes.
- (b) The sample count doubles; bits per sample stay fixed. Find the new payload in bits.
- (c) Explain why this change does not guarantee an exact representation of every possible analog signal.

8. [4 marks]

A 1,200-byte digital dataset can be stored as a 900-byte lossless file or a 300-byte lossy file.

- (a) Find the bytes saved by each option.
- (b) Which option meets a requirement to recover all original records exactly? Explain.
- (c) Give a different use where the lossy option might be acceptable, and state the condition.

9. [4 marks]

The raw survey has 16 rows: one repeated response ID, one missing time, one negative time, and an unusual 120-minute time. A label is also “ bus ”.

- (a) State how to standardize the label.
- (b) Under the supplied rules, how many records remain?
- (c) Explain why 120 remains.
- (d) Name one item the analysis log should preserve.

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Written responses | Questions 10-12

10. [4 marks]

Baseline: 13 retained times sum to 381 minutes and have median 20. A comparison omits the 120-minute record.

- (a) Calculate the baseline mean to two decimal places.
- (b) Calculate the comparison count, sum and mean.
- (c) Does the changed mean prove the omitted record is wrong? Explain.

11. [4 marks]

The baseline means are Walk 11.25 minutes (n=4), Bike 17.67 (n=3) and Bus 47.17 (n=6).

- (a) Choose a chart suitable for comparing the means and state its category and value labels.
- (b) Give one caption detail needed to identify the analysis view.
- (c) Explain why the chart does not show every individual travel time.

12. [4 marks]

A student concludes, "Bus travel causes long commutes for all students." The synthetic voluntary survey covers one school on one morning; mode groups also differ in distance.

- (a) Rewrite the claim using the baseline group statistics.
- (b) Give one numerical piece of evidence with units and n.
- (c) Explain one causal limitation and one population limitation.
